- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

9 A car travelling in a straight line accelerates uniformly from rest to $V\,\mathrm{m\,s^{-1}}$ in Ts. It then slows down uniformly, coming to rest after a further $2T$s.

(a) Sketch a velocity-time graph for the car. **[2]**

The acceleration in the first stage of the motion is $2.5\,\mathrm{m\,s^{-2}}$ and the total distance travelled is 240 m.

(b) Calculate the values of V and T. **[4]**